

Obul Reddy Lekkala

Data Engineer

Redmond, WA | obulreddy.le@gmail.com | (774)737-4255 | [LinkedIn](#)

PROFESSIONAL SUMMARY

Data Engineer with 6+ years of experience designing and building **scalable data pipelines** across **cloud and on-premise platforms**. Strong expertise in **Apache Spark (PySpark)**, **Apache Kafka** and **ETL pipeline development**, processing **high-volume datasets** and **real-time streaming data**. Experienced in building **AWS-native data platforms** using **AWS Glue**, **Amazon Redshift**, **S3** and **Databricks**, supporting **analytics**, **BI** and **forecasting workloads**. Proven ability to **optimize data pipelines** and **ETL performance**, reducing **data latency by 70%** while implementing **data quality frameworks** and **automated validation checks**. Skilled in **dimensional modeling**, **event-driven architectures** and **CI/CD automation**, delivering **production-grade data solutions** for **manufacturing analytics**, **inventory forecasting** and **revenue planning systems**.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL (PL/SQL, T-SQL), Linux/Unix Shell Scripting

Big Data & Distributed Processing: Apache Spark (PySpark, Spark SQL), Hadoop (HDFS, YARN, Hive), Apache Kafka, Parquet, ORC

ETL & Data Orchestration: Apache Airflow, AWS Glue, Informatica PowerCenter, Control-M, Batch Processing, Incremental Data Loading

Databases & Data Warehousing: Amazon Redshift, Oracle, SQL Server, PostgreSQL, Dimensional Modeling, Query Optimization, Performance Tuning

Cloud Platforms: AWS (EC2, S3, Glue, Redshift, SQS, SNS, EKS, CloudWatch)

DevOps & CI/CD: Docker, Kubernetes (Amazon EKS), Jenkins, Git, GitHub Actions, CI/CD Pipeline Automation

Data Processing & Analytics: Databricks, Pandas, NumPy, Data Quality Validation, Schema Design, Data Profiling, Data Cleansing

Tools & Platforms: JupyterHub, Kafka Connect, SFTP, systemd, Gerrit Code Review, Confluence

Core Competencies: Data Pipeline Development, Real-Time Data Processing, Event-Driven Architecture, Streaming Data Pipelines, Data Quality Frameworks, Performance Optimization, Root Cause Analysis, Agile/Scrum

EXPERIENCE

CDS Global – Data Engineer

Location: Remote (WA), USA

Duration: February 2024 – Present

- Develop **scalable data pipelines** using **AWS Glue**, **Databricks**, **Apache Spark** and **PySpark** to process **large-scale transactional and time-series datasets**, reducing **data-to-consumption latency by 35%** for **inventory forecasting** and **revenue analytics**.
- Build **PySpark transformations** and **SQL pipelines** implementing incremental data loading, schema validation and data quality checks, improving **data freshness and reliability** for downstream **analytics** and **BI workloads**.
- Implement **GPU-accelerated batch processing workflows** on **AWS EC2**, optimizing compute-intensive feature engineering pipelines and reducing **processing time from 6 minutes to 1.5 minutes per batch**.
- Design **event-driven data ingestion pipelines** using **AWS SQS**, **SNS** and **Amazon S3**, enabling **near real-time processing of transactional events** and loading curated datasets into **Amazon Redshift** for reporting and forecasting use cases.
- Integrated **AI-powered demand forecasting models** into the **data pipeline** using **Amazon SageMaker** and **Python**, automating the generation of **predictive inventory insights** that improved forecast accuracy by **20%** and reduced stockouts by **15%** for key retail clients.
- Develop **containerized data services** using **Docker** and deploy to **Amazon EKS (Kubernetes)** with **auto-scaling** and **health monitoring**, maintaining **99% pipeline uptime** under variable workload demands.

Intel – Data Engineer

Location: India

Duration: August 2019 – November 2022

- Developed **real-time data pipelines** using **PySpark Structured Streaming** and **Apache Kafka**, processing **8-12 GB of daily equipment telemetry** at **1-10 second intervals** and storing validated datasets in **HDFS (Parquet)** for downstream analytics.
- Built and maintained **Apache Airflow DAGs** orchestrating **batch and streaming ETL workflows**, including sensor data normalization, equipment metadata enrichment and statistical aggregation, reducing raw telemetry **volume by 95%**.
- Implemented **data quality validation frameworks** for **50+ sensor parameters**, introducing **anomaly detection** and **quarantine routing** that helped identify **sensor calibration drift** in semiconductor manufacturing equipment.
- Optimized **Apache Spark batch processing jobs** using **partition pruning** and **broadcast joins**, reducing **data transformation runtime from 40 minutes to 12 minutes (70% improvement)** for lot-level analytics workloads.
- Designed **integrated manufacturing analytics pipelines** combining **wafer test data**, **equipment telemetry**, **process history** and **defect inspection datasets**, producing **lot-centric analytical tables** supporting engineering yield analysis.
- Executed **large-scale data backfills** using **Apache Airflow**, loading **2B+ sensor records** and **600M+ wafer test records**, while maintaining **CI/CD pipelines** (Jenkins, Docker, Kubernetes) for **PySpark pipeline deployment**.

Capgemini – ETL Developer

Location: India

Duration: June 2018 – August 2019

- Developed **ETL pipelines** using **Informatica PowerCenter** to integrate **Oracle** and **SQL Server** data into an **Oracle 12c Data Warehouse**, processing **30M+ semiconductor manufacturing records**.
- Implemented **incremental data loading strategies** (**high-watermark extraction**) and optimized **lookup transformations** and **caching**, reducing **ETL pipeline runtime from 2 hours to 45 minutes (65% performance improvement)** while meeting **2-hour SLA refresh cycles**.
- Built **sensor data aggregation pipelines** processing **hundreds of thousands of equipment readings**, applying **statistical transformations** (**mean, std dev, min/max**) that reduced raw data volume by **95%** and improved **equipment performance monitoring**.
- Developed **data quality and analytics data marts** with data profiling, cleansing rules and cross-system harmonization, enabling **yield excursion analysis** via **JMP/ODBC** and reducing **root-cause analysis time from 24 hours to under 5 hours** for manufacturing defects.

EDUCATION

Master's in Data Science, University of Massachusetts, Dartmouth, MA, USA.